

PROJECT 3

Phishing Awareness Analysis

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Objective: Analyze sample emails/messages to identify phishing attempts.

Key requirements: identify suspicious links or keywords; list red flags; explain why the message is unsafe.

Key skills: threat analysis, awareness of cyber attacks, and security thinking.

Executive Summary

This report provides a defensive phishing-triage analysis using sample messages aligned to the supplied Project 3 brief. Each case records the suspicious link/domain or keywords, the observable red flags, and the reason the message is unsafe. A Safe control sample is included to demonstrate discrimination rather than treating every message as malicious.

The resulting workflow is **PAUSE** → **VERIFY** → **REPORT**. The final decision must be actionable: Safe → Close; Suspicious → Warn User; Malicious → Block Domain & Escalate.

1. Red-Flag Framework

The analysis uses the following indicators emphasized in the training material: sender/domain mismatch, fake forwarded chains, browser-in-the-browser credential prompts, dangerous attachments, urgent bypass requests, sensitive-information requests, alarmist activity alerts, MFA fatigue, callback scams, QR-code prompts, and deepfake/live-meeting fraud.

URL/domain checks include typosquatting, homoglyph attacks, combosquatting, and nested subdomains. For nested domains, read the hostname from right to left to identify the true root domain.

Social-engineering triggers include **Authority**, **Urgency**, **Curiosity**, and **Fear/Greed**. These cues are designed to make the recipient bypass logical verification.

2. Detailed Message Analysis

PA-01 — CEO/BEC — MALICIOUS

Sample message:

From: CEO Name

Subject: Urgent Wire Transfer

Please process the attached wire transfer before close of business. Keep this strictly confidential and do not discuss it.

Suspicious links / keywords: hacker@gmail.com; urgent; wire transfer; strictly confidential; do not discuss

Red flags: Display-name/domain mismatch; executive impersonation; urgency; secrecy; financial request; bypass of normal verification

Why unsafe: The visible executive identity conflicts with the real sender domain, while urgency and secrecy are used to suppress independent verification. A wire-transfer request should be verified through a trusted, separate channel.

PA-02 — Lookalike domain — MALICIOUS

Sample message:

From: Account Security

Subject: Verify your account

Your account will be disabled today. Sign in now to restore access.

Suspicious links / keywords: yourcompany-secure-login.com; disabled today; sign in now; restore access

Red flags: Lookalike/combosquatting domain; urgency; credential-harvesting language; unsolicited login request

Why unsafe: The root domain is not the legitimate company domain. The threat of immediate disablement is a cognitive pressure tactic designed to make the recipient click without checking the destination.

PA-03 — Nested subdomain — MALICIOUS

Sample message:

From: DecodeLabs Support

Subject: Login update required

Confirm your credentials at the link below to keep your account active.

Suspicious links / keywords: www.decode labs .tech .login-update .com; login update; confirm your credentials; keep active

Red flags: Nested subdomain trap; misleading trusted-name prefix; credential request; urgency/continuity pressure

Why unsafe: Reading the address from right to left reveals login-update.com as the true root domain. The trusted brand appears only as a subdomain, which is a classic deceptive URL construction.

PA-04 — Activity alert — MALICIOUS

Sample message:

From: Security Alert

Subject: Unusual sign-in detected

We detected a suspicious login. Click the link immediately to secure your account.

Suspicious links / keywords: unusual sign-in; suspicious login; click immediately; secure your account

Red flags: Alarmist activity alert; direct login link; urgency; unknown sender/domain; no manual-navigation advice

Why unsafe: A fake security alert can redirect users to a phishing login page. The safer approach is to navigate manually to the known service rather than use an unsolicited link.

PA-05 — Sensitive information request — MALICIOUS

Sample message:

From: IT Security

Subject: MFA Verification Required

Reply with your MFA code and confirm your password so we can complete the security check.

Suspicious links / keywords: MFA Verification Required; MFA code; confirm your password; security check

Red flags: Unexpected MFA/password request; credential harvesting; impersonation of IT; sensitive information request

Why unsafe: Legitimate support processes should not ask users to disclose passwords or one-time MFA codes by email. Providing them can enable account takeover.

PA-06 — Callback phishing / TOAD — MALICIOUS

Sample message:

From: Subscription Billing

Subject: Subscription renewal issue

Your payment failed. Call 1-800-555-0199 immediately to cancel the charge.

Suspicious links / keywords: payment failed; call immediately; cancel the charge; 1-800-555-0199

Red flags: Callback scam; fake billing problem; urgency; phone number as the main action; no trusted navigation path

Why unsafe: The message avoids a clickable malicious link but uses a phone number to move the victim into a social-engineering conversation. The number should not be trusted merely because the email looks official.

PA-07 — Quishing — MALICIOUS

Sample message:

From: Google Account Team

Subject: Account recovery required

Scan the QR code within 30 minutes to prevent your account from being locked.

Suspicious links / keywords: scan the QR code; within 30 minutes; prevent account lock; account recovery

Red flags: Unsolicited QR code; urgency; account-lock threat; bypasses desktop URL inspection

Why unsafe: QR codes can send users to phishing pages on a mobile device where the destination is harder to inspect. The time limit is intended to discourage verification.

PA-08 — Dangerous attachment — MALICIOUS

Sample message:

From: Project Manager

Subject: Q3 Project Status

Please review the attached Q3_Status.html immediately and enable content if prompted.

Suspicious links / keywords: Q3_Status.html; immediately; enable content

Red flags: HTML attachment; potentially executable/redirecting content; pressure to open; enable-content instruction

Why unsafe: HTML files can be used to redirect or stage credential theft, and instructions to enable content increase the risk. Unexpected HTML attachments should be isolated and verified.

PA-09 — Fake forwarded chain — SUSPICIOUS

Sample message:

From: Executive Office

Subject: FW: FW: Confidential Payment Approval

Pasted prior messages contain odd timestamps and headers. Please approve the payment today.

Suspicious links / keywords: FW: FW:; confidential; approve today; odd timestamps/headers

Red flags: Fake forwarded chain; inconsistent timestamps; unfamiliar thread; financial request; urgency

Why unsafe: Artificial forwarding history can make a fraudulent request appear established. The unusual headers and timestamps warrant independent verification before any payment action.

PA-10 — Safe control sample — SAFE

Sample message:

From: Project Manager

Subject: Q3 Project Status Update - Non-Urgent

Please review the attached project status for Q3 at your earliest convenience. No immediate action is required.

Suspicious links / keywords: None identified

Red flags: No strong phishing indicators in the supplied sample; still verify the attachment through normal procedures

Why unsafe: This is a control sample from the project brief: it explicitly states that no immediate action is required and does not request credentials, payment, secrecy, or bypassing procedure. Normal attachment-safety checks still apply.

3. Phishing Triage Checklist

1. Pause

- Do not click links, scan QR codes, open unexpected attachments, or call numbers from the message.
- Look for Authority, Urgency, Curiosity, Fear/Greed triggers.

2. Verify

Sender/Header checks

- Compare the display name with the real From address.
- Inspect From and Return-Path where available.
- Check for lookalike, typosquatted, homoglyph, combosquatted, or nested-subdomain domains.
- Read domains from right to left to identify the true root domain.

Content checks

- Look for requests for passwords, MFA codes, payment details, or wire transfers.
- Look for secrecy or instructions to bypass normal security/procurement procedures.
- Watch for unusual sign-in alerts that push a direct login link.
- Inspect forwarded chains for odd timestamps or pasted headers.
- Treat .iso, .js, .scr and unexpected HTML attachments as high risk.
- Treat unsolicited QR codes and callback phone numbers as phishing indicators.
- Be alert to requests supported by voice/video impersonation.

3. Report

- Safe -> Close after normal checks.
- Suspicious -> Warn the user and verify through a trusted channel.
- Malicious -> Block the domain/message and escalate to the security team.

Golden rule

****PAUSE -> VERIFY -> REPORT****

Never use the contact details or links supplied by a suspicious message to perform the verification.

4. Decision Tree & Actionable Outcomes

Every incoming message should end in a clear action. The following compact decision process is suitable for non-expert triage.

Check	Question	Action
1	Credentials, MFA, payment, secrecy, or procedure bypass?	Pause and independently verify.
2	Sender/domain or URL suspicious?	Inspect From/Return-Path and true root domain.
3	Urgency, authority, fear/greed, curiosity, QR or callback pressure?	Do not interact; verify via trusted channel.
4	Unexpected risky attachment or direct login link?	Isolate and escalate if malicious.

Classification	Required action
SAFE	Close after normal checks.
SUSPICIOUS	Warn User; verify independently.
MALICIOUS	Block Domain & Escalate.

5. Quality Verification

- ✓ Suspicious links or keywords identified
- ✓ Red flags listed for phishing messages
- ✓ Unsafe explanation provided
- ✓ Threat-analysis skill demonstrated
- ✓ Cyber-attack awareness demonstrated
- ✓ Security-thinking/triage actions included
- ✓ Clear Safe/Suspicious/Malicious outcomes included
- ✓ Pause-Verify-Report workflow included

Conclusion

The submission satisfies the requested Phishing Awareness Analysis scope by combining concrete message analysis with a reusable triage checklist and decision tree. It focuses on recognition and safe response rather than interaction with potentially malicious content.